

Co-Pilot AI Scientist v3: Insight-Gated Research Evolution for Collaborative Automated Science

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Abstract

Autonomous research agents have begun to connect idea generation, experiment execution, benchmark evaluation, and paper writing. Yet fully autonomous pipelines still struggle at the creative and strategic decisions where human scientists contribute taste, field knowledge, and responsibility for claims. We propose Co-Pilot AI Scientist v3 and its underlying method, Insight-Gated Research Evolution (IGRE). IGRE is not a direct composition of prior research agents. It reorganizes their useful lessons into a tail-seeking research process: automated systems maintain broad executable frontiers, while humans inject scientific taste, risk tolerance, and claim responsibility at explicit decision gates. The primary goal is not to prove a binary claim that "human participation improves paper quality." It is to design, compare, and refine human-participation modes that better translate hard-to-quantify scientific taste and insight into automated research workflows. We define a reproducible evaluation protocol that compares autonomous, partially gated, and fully collaborative variants on small automated-research tasks. The central hypothesis is intentionally asymmetric: human input may reduce average short-budget benchmark performance, but a well-designed participation workflow can raise the probability of rare, high-novelty, high-impact research outcomes that matter most for scientific discovery.

1. Introduction

The next useful version of automated research is unlikely to be a scientist-free paper factory. Scientific work is not only a sequence of executable steps; it is also a process of choosing valuable questions, recognizing weak evidence, reframing failures, and deciding which claims deserve to be made. Prior systems offer useful ingredients: hypothesis debate, executable experiment search, and automatic-evaluator code evolution. This paper uses those ingredients as inspiration, but the method is defined around a different objective. Scientific co-piloting should not only maximize the mean score of short benchmark runs. It should also increase the chance that a run crosses into a qualitatively better research direction: a sharper question, a more revealing evaluator, a surprising failure interpretation, or a claim that opens a new line of work.

This paper proposes Co-Pilot AI Scientist v3 and formalizes it as Insight-Gated Research Evolution (IGRE). The goal is not to slow an autonomous pipeline with arbitrary approvals. IGRE treats human intervention as a scarce, high-variance operator that should be invoked only where scientific judgment can change the shape of the search frontier. The proposed gates are: scientific taste prior, evaluator stress test, frontier steering, verifiable micro-evolution, and claim calibration.

The practical question is therefore a design question: which human participation pattern best converts scientific taste into useful search pressure? IGRE evaluates several modes rather than assuming a single human role: upstream taste-prior selection, evaluator stress testing, mid-run frontier steering, selective program-search escalation, structured feedback on manuscript evidence, and claim calibration. The experiments in this version are used to shape that workflow and identify boundary conditions, including negative cases where autonomous search is stronger under short budgets. This makes the paper an applied collaboration-design study for automated science, not merely a test of whether humans are globally "better" or "worse" than automation.

2. Related Work

AI Co-Scientist frames scientific discovery as hypothesis generation, discussion, and evolution. Its strength is upstream research imagination and evidence organization. AI Scientist-v2 focuses on turning candidate ideas into runnable experiments and papers through agentic tree search. AlphaEvolve focuses on code evolution driven by automatic evaluation. FunSearch provides an earlier example of LLM-guided program search for mathematical discovery. Coscientist shows how LLM agents can connect to chemistry tools and laboratory automation.

IGRE borrows design pressure from these systems, not their control logic. From hypothesis agents it takes the need for diverse conjectures, but replaces free-form debate with a logged scientific-taste prior. From automated paper agents it takes executable experiment search, but inserts frontier steering and claim calibration where benchmark score is not enough. From program evolution it takes deep search on machine-gradeable subproblems, but exposes that search only through a selective escalation gate. The result is a co-pilot pattern whose unit of optimization is not a single task metric, but an evidence-aligned research trajectory with higher upside.

2.1 Distinction from Generic Research Co-Pilots

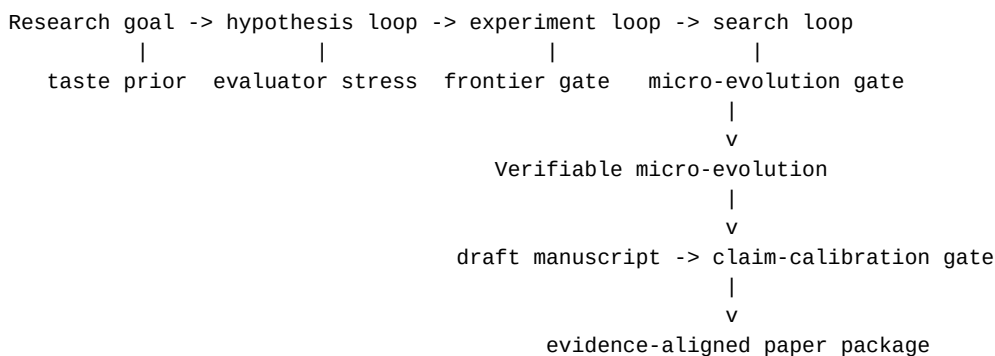
Many co-pilot systems treat humans as approvers, prompt writers, preference labelers, or emergency supervisors. IGRE makes a different algorithmic claim: the most important human contribution to automated science is often a non-metric prior over what is scientifically tasteful. This prior is difficult to reduce to a scalar reward. It includes judgments such as whether a question has depth, whether a negative result is revealing, whether a benchmark is too easy to game, whether a surprising branch is worth preserving, and whether a claim would still matter if the primary metric improved only slightly.

IGRE therefore separates three concepts that are often conflated in human-in-the-loop agents. Preference means choosing what looks better among finished options. Oversight means preventing invalid or unsafe actions. Scientific taste means changing what the system searches for before the outcome is known. The last role is the distinctive one. In IGRE, taste is not forced into a complete reward model; it is captured as a structured, auditable-but-partly-qualitative gate record that changes the search frontier. This lets the system evaluate human input honestly: it may hurt the mean score, but it should be judged by whether it increases the chance of rare, field-opening trajectories.

3. Method

Co-Pilot AI Scientist v3 implements Insight-Gated Research Evolution. IGRE has four machine loops and five human-facing operators. The machine loops preserve the scalability of automated research: hypothesis expansion, experiment construction, executable frontier search, and verifiable micro-evolution. The human-facing operators are designed for places where scientific taste can alter which frontier is worth expanding.

Figure 1 gives the operational data flow. A research goal enters the hypothesis loop, passes through experiment construction and executable frontier search, optionally delegates machine-gradeable subproblems to OpenEvolve-based micro-evolution, and finally reaches manuscript generation and claim calibration. Each gate consumes a structured frontier and produces a logged decision artifact.



First, the scientific-taste prior asks the human scientist to select, merge, or rewrite candidate directions using criteria that are not fully captured by early metrics: conceptual freshness, field relevance, asymmetry of upside, and whether a failure would still teach something.

Second, the evaluator stress-test gate converts selected hypotheses into benchmark tasks, baselines, ablations, executable scripts, and rejection conditions. The gate asks not only whether the metric is convenient, but whether it can be gamed, whether it misses utility, and whether it would support the paper claim if the result were positive.

Third, the frontier-steering gate runs over executable experiment branches. At predefined checkpoints, the system summarizes metrics, code snapshots, errors, and novelty notes. The human scientist may allocate budget to a branch that is not currently best on the primary metric if it has higher scientific upside or reveals a more important failure mode.

Fourth, the verifiable micro-evolution gate sends only machine-gradeable subproblems to a code-evolution engine. Since AlphaEvolve itself is not open sourced, our reproducible implementation uses OpenEvolve as the practical substrate for this operator. OpenEvolve provides an evolutionary coding loop with custom evaluators, OpenAI-compatible model routing, MAP-Elites quality-diversity search, island-based populations, and reproducible seeds. The operator evolves code, stores candidate programs, and returns verified improvements to the main research pipeline.

Fifth, the claim-calibration gate audits the final paper draft against the evidence record. It weakens, removes, or reframes claims that are not supported, and it records which stronger claims remain hypotheses for the next benchmark stage.

Algorithm 1: Insight-Gated Research Evolution (IGRE)

Input: research goal g , benchmark budget B , human attention budget H

1. Generate a diverse hypothesis frontier F_h from g .
2. Apply `scientific_taste_prior( $F_h$ ,  $H$ )` to select or rewrite high-upside hypotheses, including branches whose value is not yet metric-visible.
3. Convert selected hypotheses into evaluators, baselines, and failure tests.
4. Apply `evaluator_stress_test` to reject metrics that are non-executable, easily gamed, or too weak to support the intended claim.
5. Run executable frontier search under budget B and log branch states.
6. At checkpoints, apply `frontier_steering` to allocate budget using both metrics and qualitative upside/failure value.
7. For machine-gradeable subproblems, apply `verifiable_micro_evolution` only when direct editing is insufficient and the evaluator is reliable.
8. Draft the paper from logs, then apply `claim_calibration` to align claims with evidence and demote unsupported conclusions to future hypotheses.

Output: evidence-aligned paper package plus auditable gate trajectory.

Each human intervention is recorded as structured data: decision type, options, free-text rationale, affected artifacts, and downstream consequences. This makes human participation part of the reproducibility record rather than an invisible side channel.

The current artifact package contains a retrospective full-gate trajectory covering all proposed gate types. It links an idea gate selecting the narrow human-gated tree-search contribution, an evaluator gate rejecting a fairness metric-gaming branch, a live branch gate selecting among two Causality drafts, a program-search gate escalating to OpenEvolve, and a claim gate demoting unsupported superiority claims to future work. This trajectory demonstrates the log format and evidence chain, but it is not yet a single online end-to-end run.

We also provide a rerunnable full-gate trace script. Given the archived experiment summaries, the script recomputes a continuous sequence of idea, evaluator, branch, program-search, and claim gates and writes both JSON and Markdown artifacts. This verifies that the gate policy can be executed over the current evidence package. It is deliberately labeled as an executable artifact replay rather than a fresh online training run.

After that replay artifact, we ran a first fresh online full-gate smoke trajectory on ubuntu-heshi. The orchestrator generated an idea gate, approved a low-cost evaluator bundle, launched a new two-draft FML-bench Causality frontier, selected the lower-validation-MAE branch, continued the selected snapshot for one AI Scientist-v2 step, ran a one-iteration OpenEvolve knapsack search, and wrote a claim-audit gate. The branch frontier selected step 2 (0.627837 validation MAE) over step 1 (0.677448). The one-step continuation then reached test MAE 0.862015, worse than the selected frontier's test MAE 0.646224; the simultaneous knapsack OpenEvolve smoke reached best score 1.000000. This is the first online orchestration evidence for all five gate types, but it is a smoke test and a negative continuation outcome, not evidence of co-pilot superiority.

We then ran a same-FML-step autonomous baseline for the online smoke. The baseline used the same Causality task and DeepSeek model for three AI Scientist-v2 steps, but without a human branch gate. It reached validation MAE 0.354147 and held-out test MAE 0.428516, substantially outperforming the human-gated smoke continuation's test MAE 0.862015. This paired smoke result is negative for performance improvement, while still supporting the narrower claim that online co-pilot orchestration is executable.

We subsequently repeated the stronger same-continuous-trajectory paired online smoke pattern three times. Each run generated complete co-pilot and autonomous manuscript artifacts from the same orchestrator invocation. Two Causality runs have valid scalar test metrics: the co-pilot continuation has mean test MAE 0.754120 and the same-run autonomous baseline has mean test MAE 0.643337; lower is better, so the valid-metric aggregate is 0 co-pilot wins, 1 autonomous win, and 1 tie. The third run switches to Fairness_fairlearn and is archived as a no-valid-branch failure-mode trajectory; neither side obtains a valid scalar test metric. Monica-routed A/B model-review probes prefer the co-pilot manuscript in all six reviewer calls across the three runs. This is stronger than a single archived matched-budget comparator because the manuscripts come from repeated same-run online smokes, including a cross-task failure case. It still does not prove co-pilot superiority: the valid benchmark aggregate favors autonomous or tie, the budgets are tiny, and model review is not independent expert paper-quality review.

We now also include a live AI Co-Scientist-style hypothesis-frontier smoke. A Monica-routed gpt-4o-mini front end made two live model calls: one to generate four candidate research frontiers for the next IGRE evidence milestone, and one to critique and rank them under conservative AI Scientist-v2 evidence discipline. The selected next-budget candidate is frontier_004, a structured human-feedback mechanism intended to compare rubric-guided feedback against informal feedback for reproducibility and clarity. This closes a narrow orchestration gap in the AI Co-Scientist portion of the architecture: the package now has a logged generate-critique-rank hypothesis front end. It does not show that the selected candidate improves downstream benchmarks or paper quality, and it is not a human-selection result.

We then added an autonomous hypothesis-front-end baseline. The script `run_hypothesis_frontend_baseline.py` reuses the archived IGRE candidate portfolio, asks the same model to generate an autonomous AI Scientist-v2-style portfolio without explicit human-taste gates or structured human feedback, and uses a fixed-rubric comparison call to score both portfolios. The scorer prefers IGRE (overall 4 versus 3) because the IGRE portfolio has higher evidence gain, benchmark fit, claim calibration, and human-taste visibility. The autonomous portfolio is not useless: the audit marks it as stronger for exploring fully autonomous process variants. This probe only compares direction-finding portfolios; it does not evaluate downstream benchmark performance, paper quality, or human expert judgment.

We therefore ran a small downstream structured-feedback probe for frontier_004. Starting from the same archived co-pilot manuscript, the script `run_structured_feedback_probe.py` made five Monica-routed gpt-4o-mini calls: free-form informal feedback, IGRE-structured feedback, one revision conditioned on each feedback mode, and a fixed-rubric model score over the two revisions. The scorer preferred the structured-feedback revision (overall 5 versus 4), with higher scores for clarity, reproducibility, claim calibration, evidence grounding, method distinctness, limitation honesty, and novelty preservation. This result should be read narrowly. It shows that the IGRE feedback format can be operationalized and can induce a measurable revision difference under a model-routed evaluator. It is not independent human peer review, not a multi-task result, and not evidence that human feedback improves benchmark performance.

To test whether public expert-review data can serve as an offline proxy for scientific taste, we added an expert-review taste-prior probe over nhop/OpenReview. The script `run_expert_review_taste_prior_probe.py` first checks Hugging Face Dataset Viewer availability, then uses streaming access so the experiment does not need to download the full dataset. In the current run, the dataset statistics report 34,638 rows and the streaming sample contains 160 papers. The required fields are available, including titles, abstracts, reviews, decisions, mean overall score, novelty, clarity, impact, and reproducibility fields. Within the 160-row sample, `mean_score` is available for all 160 rows with mean 0.5349, `mean_novelty` for 27 rows with mean 0.6284, `mean_correctness` for 106 rows with mean 0.6267, `mean_clarity` for 71 rows with mean 0.5968, `mean_impact` for 73 rows with mean 0.5459, and `mean_confidence` for 158 rows with mean 0.6373. No sampled row has a usable `mean_reproducibility` value, which is an important caveat for any reproducibility-aware taste model. This probe supports only a limited offline taste-prior construction: it can help rank or diagnose generated hypotheses and manuscript revisions against expert-review signals, but it is not live human co-pilot interaction data and does not show that such a prior improves AI Scientist-v2 outputs.

This placement makes the OpenReview data useful for the paper's main design question. Rather than treating the dataset as evidence that online human co-piloting has already occurred, we use it as a participation-mode selection benchmark: different workflow variants can generate hypotheses, experiment plans, evidence summaries, or manuscript revisions, and an OpenReview-derived rubric can score whether those artifacts resemble higher-rated expert-reviewed papers in novelty, correctness, clarity, impact, and reviewer confidence. This lets the paper compare participation patterns such as no human gate, upstream taste-prior gate, evaluator-stress gate, structured-feedback gate, and claim calibration gate under a common offline expert-review proxy. The result would not prove real-world peer-review acceptance, but it can provide data-driven support for choosing a better human-in-the-loop workflow before larger online studies exist.

4. Benchmark Selection and Evaluation Plan

We evaluate six variants: autonomous baseline, idea gate only, branch gate only, evaluator gate only, claim gate only, and full co-pilot v3. Candidate tasks should be selected by claim type rather than by a single favorite benchmark. For AI Scientist-v2-style experiment search, FML-bench is a useful near-term benchmark because it is already runnable in our Ubuntu environment and exposes branch-level logs. For machine-gradeable algorithmic subproblems, we use OpenEvolve-controlled tasks such as function minimization, knapsack heuristic search, and weighted Max-Cut. For broader evidence beyond FML-bench, we include an initial MAgentBench vectorization probe for end-to-end ML experimentation and track setup probes for additional MAgentBench and ScienceAgentBench tasks. The current evidence still does not include a second scored official non-FML benchmark: the MAgentBench CIFAR10/debug run was blocked by slow dataset download, the MAgentBench IMDB probe repaired the missing datasets dependency but could not reach HuggingFace from the Ubuntu host, and ScienceAgentBench metadata/artifacts were not reachable. Higher-cost stretch benchmarks include MLE-bench Lite for Kaggle-style ML engineering, PaperBench for paper-to-code replication and hierarchical rubric grading, and AIRS-Bench for full ML research lifecycle evaluation. Metrics include task score, paper quality, claim support, search efficiency, human attention cost, and hypothesis diversity. For the programmatic-search module, the key ablation is OpenEvolve versus repeated direct LLM edits under the same evaluator and iteration budget. FML-bench should therefore be read as one current evidence source, not as the complete benchmark definition for the project.

Because IGRE's key contribution is scientific taste rather than generic approval, we add a taste/insight rubric to the evaluation package. The rubric scores problem depth, novelty potential, mechanistic value, failure informativeness, benchmark taste, claim significance, and risk asymmetry from 1 to 5, with short rationales. This is deliberately not a reward model. It is a structured record of the non-metric prior that caused a human to preserve, rewrite, or prune a branch. Future evaluations should report both mean task performance and high-tail signals: whether a human-selected branch that an autonomous policy would have pruned later produces a stronger claim, better evaluator, or more informative negative result.

We also add a taste/insight coverage audit over 39 gate records. It finds 2 complete taste_insight records: one scientific-taste prior encoding the author's decision to move from an FML-centric benchmark story to a claim-matched benchmark portfolio and high-tail evaluation, and one operator-recorded priority gate choosing to measure attention/taste before making stronger human-efficiency or taste-effect claims. The other 37 records lack complete taste/insight fields. This is not a performance result. It does, however, make the paper's distinctive human-taste claim operational: prospective matched-budget runs must fill both taste_insight and attention_cost before IGRE can test its high-tail hypothesis.

Because the central claim concerns human scientific taste in a co-pilot research loop, the paper also needs human co-pilot usage data. We therefore add a derived Human Co-Pilot Trace Dataset protocol. A survey of public interaction datasets finds useful adjacent resources, including CoAuthor for collaborative writing, CUPID for contextual preference interactions, neulab/agent-data-collection for broad agent trajectories, and WebChain for human-annotated web trajectories. None of these directly contains human scientist interventions inside an AI Scientist-v2-style loop with linked hypotheses, benchmark runs, code artifacts, claim audits, and manuscripts. The primary dataset for this paper is therefore a single-author longitudinal Codex co-pilot trace corpus derived from this project: gate records, artifact paths, commit IDs, benchmark metrics, manuscript revisions, and claim-audit outcomes, without releasing raw chat logs or credentials. The current derived snapshot indexes 52 gate records, 35 records with attention-cost fields, 10 records with taste/insight fields, 4 prospective matched packages, and 58 relevant commits. This supports ecological and process claims, but not population-level claims about all scientists.

We add a release-readiness audit for this dataset. The audit verifies required top-level fields, gate schema coverage, explicit claim boundaries, and scans for secret-like strings and raw-log markers. The current audit passes with 4 public-dataset survey entries, 52 gate records, 4 prospective packages, 58 commit-index entries, 0 secret-pattern hits, and 0 raw-log marker hits. This permits release as a derived metadata case-study artifact, not as raw chat logs or human-subject population data.

Following the latest paper-quality review, we made attention cost an explicit auditable artifact rather than an informal metric. The human-gate schema now contains an optional `attention_cost` object with active review minutes, wall-clock latency, number of options reviewed, artifacts reviewed, and decision count. We also added a strict coverage audit over 39 gate records: 9 standalone human-gate logs and 30 embedded gates across 6 trajectory artifacts. The audit finds 1 complete attention-cost record: an operator-recorded measurement-readiness gate with 8.5 active review minutes, 9.67 wall-clock minutes, 3 options reviewed, 4 artifacts reviewed, and 1 decision. The other 38 records still lack one or more required fields. This is an important measurement-readiness result: the archived gates show decision provenance and the logging path now works, but the package still cannot support any efficiency claim about human attention. Future prospective matched runs must fill this field before comparing co-pilot and autonomous variants.

We further add a prospective matched-budget package audit. This audit is a hard evidence-shape gate rather than a result. A package passes only if it contains a prospective co-pilot trajectory, a matched autonomous baseline on the same task/model/tool budget, complete `attention_cost` and `taste_insight` records for every human gate, a final claim audit, and a manuscript generated from the same run. The repository now includes one passing controlled micro-pilot package on a weighted Max-Cut task. This proves that the package shape can be generated and audited on a real remote computation, but it is not top-conference performance evidence: it does not use AI Scientist-v2 tree search and does not evaluate paper quality.

We then upgraded this evidence shape to an AI Scientist-v2-style FML-bench pilot. The new package runs fresh `Causality_causalm1` co-pilot and autonomous baselines under the same DeepSeek model and two-step budget, records complete attention/taste fields for the human frontier gate, and writes a claim audit plus mini-manuscript. The result is negative for co-pilot performance in this small run: co-pilot test MAE is 0.646224, while the matched autonomous run reaches 0.624703 (lower is better). This is valuable because it verifies the prospective evidence package on an AI Scientist-v2-style task while reinforcing that superiority remains unproven.

We additionally report a metric-level prospective package summary rather than only a pass/fail artifact audit. It currently contains three passing packages: one controlled Max-Cut micro-task in which the human-selected branch wins on mean normalized score (0.984419 versus 0.596214), and two FML-bench Causality packages in which the autonomous run wins on test MAE (0.624703 versus 0.646224, and 0.296399 versus 0.646224). All three human gates have complete attention-cost and taste/insight records. This summary is central to the IGRE framing: human scientific taste is a logged, high-variance intervention whose value must be tested across a distribution of research trajectories, not assumed from a single favorable case.

Finally, we add a first matched mini-manuscript quality probe for the FML-bench prospective package. The probe generates an autonomous mini-manuscript from the same baseline evidence, anonymizes the co-pilot package manuscript as A and the autonomous manuscript as B, and asks Monica-routed gpt-4o-mini and claude-3-7-sonnet-latest to score claim calibration, evidence use, methodological completeness, limitation honesty, clarity, and overall quality. Both reviewers prefer A, with overall scores 4 versus 3. The rationales credit the co-pilot manuscript for stronger claim calibration and more explicit limitations despite its worse benchmark score. This is only a measurement-readiness result: the artifacts are short package manuscripts from one FML task, not complete end-to-end generated papers.

To make this principle operational, we maintain a benchmark-to-claim matrix. FML-bench Causality supports the branch-gate feasibility claim but does not yet show general human-gate superiority. FML-bench Fairness supports the need for evaluator gates because executable repairs and degenerate predictors reveal metric-gaming risk. OpenEvolve function minimization, knapsack, Max-Cut, MLAGentBench vectorization, and sklearn diabetes probes test the program-search escalation policy under different subproblem types. ScienceAgentBench, MLE-bench Lite, PaperBench, and AIRS-Bench remain expansion targets rather than current scored claims. Future runs should therefore be chosen by the weakest unsupported claim, not by convenience.

4.1 Preliminary Programmatic-Search Smoke Test

As an initial feasibility check, we ran OpenEvolve 0.2.27 on the SSH-controlled Ubuntu host using DeepSeek through an OpenAI-compatible API. The task was a small function-minimization evaluator with a random-search baseline. In one OpenEvolve iteration, the system improved the evaluator score from 0.0345 to 0.0378 and saved a simulated-annealing-style evolved program. This result is only a smoke test: it verifies the remote OpenEvolve path, evaluator loading, model routing, checkpointing, and artifact capture. It does not yet establish the paper-level claim that human-gated research improves final manuscript quality.

We also ran a direct one-shot LLM-edit baseline with the same DeepSeek model, initial program, and evaluator. The direct edit reached a score of 0.038021, slightly above both the one-iteration OpenEvolve score of 0.037816 and a five-iteration OpenEvolve score of 0.038007. This boundary result supports a central design choice of Co-Pilot AI Scientist v3: programmatic search should be controlled by an explicit escalation gate. For small budgets or simple local rewrites, direct editing may be sufficient; for richer search spaces and larger budgets, OpenEvolve-style population search can become the appropriate tool.

To test that latter case, we added a richer 0/1 knapsack heuristic task. The initial value/weight greedy program reached an average optimality ratio of 0.991519 across 18 deterministic instances. A direct LLM rewrite improved this to 0.995270. A five-iteration OpenEvolve run improved it further to 0.999439, with no invalid instances. The evolved program introduced local improvement via one-item and two-item removals followed by greedy refill. This provides a positive example for the escalation gate: OpenEvolve-style search can be useful when the subproblem has a richer combinatorial structure and an automatic evaluator.

We then added a second controlled combinatorial task, weighted Max-Cut, to avoid relying on one hand-built knapsack result. The starter program alternates node labels and reaches normalized score 0.734680 against a deterministic multi-start local-search reference over 16 graph instances. A direct DeepSeek rewrite improves the score to 0.962237. A five-iteration OpenEvolve run with the same model and evaluator reaches 0.970833, improving over the direct rewrite by 0.008596 with no invalid instances. The margin is small and from one seed, but it is a second positive algorithmic subproblem result. Combined with the negative function-minimization boundary case, it supports the selective program-search gate rather than an unconditional OpenEvolve policy.

4.2 Retrospective AI Scientist-v2 Branch-Gate Replay

We also replayed two existing AI Scientist-v2 FML-bench smoke runs. In the `Causality_causalm1` run, the first draft achieved the best validation MAE observed in the four-step run (0.598943 versus a baseline of 1.296259), while the second draft and later refinements failed to improve it. A branch gate after two drafts would have kept the first branch and avoided at least one low-value follow-up. In the `Fairness_fairlearn` run, the first draft worsened the baseline fairness metric (0.316467 versus baseline 0.186632), the second attempt failed with a Fairlearn compatibility issue, and later attempts were either worse or buggy. A human branch gate would have paused the branch and routed the system to an evaluator or algorithm reset. This replay evidence is not a substitute for an online human-gated benchmark, but it shows that the logged AI Scientist-v2 trajectory contains actionable checkpoints for human co-pilot intervention.

4.3 Live AI Scientist-v2 Branch-Gate Probe

To move beyond retrospective replay, we ran a new live FML-bench probe on the Ubuntu host. We configured AI Scientist-v2 on `Causality_causalm1` with a two-step budget, two draft ideas, two simulated parallel workers, and a temporary stage-budget schedule that assigns all budget to the first stage. This produced a real two-branch frontier before a branch gate. The first draft obtained validation MAE 1.149610, while the second draft obtained validation MAE 0.621262; lower is better. The structured branch gate therefore selected the second draft and pruned the first. The benchmark's automatic final test on the selected validation branch reported test MAE 0.640451.

This live probe demonstrates that the proposed branch gate can be inserted at a real AI Scientist-v2 decision frontier and can make a clear budget-allocation decision from logged metrics and code snapshots. It does not yet prove that human gating outperforms autonomous AI Scientist-v2, but it gives a concrete frontier on which such a comparison can be built.

We then implemented a minimal selected-branch continuation runner. The runner temporarily applies the human-selected code snapshot to the official FML-bench task template, launches a short AI Scientist-v2 continuation run, and restores the template afterward. Starting from the selected second draft, a two-step continuation reached validation MAE 0.401240 and test MAE 0.402170. This is better than the live two-draft gate run's test MAE of 0.640451 and also better than the earlier four-step autonomous smoke run's test MAE of 0.617719. The comparison is still preliminary because continuation is implemented by snapshot seeding rather than by preserving the original in-memory tree object, and it uses only a single small task and seed. Nevertheless, it demonstrates an executable path from branch gate, to selected snapshot, to additional AI Scientist-v2 search budget.

Following the paper-quality review, we added closer matched-budget autonomous baselines. The first matched run used the same `Causality_causalm1` task, the same DeepSeek model, two initial ideas, two parallel branches, and four total AI Scientist-v2 steps, but no human branch selection. It reached validation MAE 0.389451 and test MAE 0.421474. In this pair, the human-gated path remained slightly better on held-out test MAE (0.402170 versus 0.421474), while the autonomous baseline was slightly better on validation MAE.

We then ran a second matched Causality replicate. The human-gated two-draft frontier selected a branch with validation MAE 0.605881 and test MAE 0.646224. The snapshot-seeded continuation did not improve the held-out test score, ending with validation MAE 0.627837 and test MAE 0.646224. The matched autonomous four-step baseline reached validation MAE 0.537972 and test MAE 0.595685. In this second pair, the autonomous path was better on both validation and held-out test metrics.

Across the two matched pairs, one pair favors human-gated continuation and one favors the autonomous baseline on held-out test MAE. Mean human-gated test MAE is 0.524197, while mean autonomous test MAE is 0.508579. Since lower is better, the two-pair mean slightly favors the autonomous baseline. The evidence therefore removes one compute-budget confound but remains mixed; it supports feasibility of branch-gate insertion and continuation, not general human-gate superiority.

We now also generate this FML matched-comparison aggregate with a script rather than relying only on the hand-written table. The generated audit reports one human-gated win and one autonomous/tie win across the two formal pairs, mean autonomous-minus-human delta -0.015618, delta SEM 0.034921, and statistical claim `not_supported_n_too_small`. The separate online-smoke comparison is also negative for co-pilot performance (0.862015 versus 0.428516 test MAE). This scripted summary is the current authoritative FML performance evidence.

After this audit, we ran a second prospective two-step FML package with the same task, model, tool access, and budget shape as the first prospective package. The co-pilot branch again reached test MAE 0.646224, while the matched autonomous baseline reached 0.296399. Across the two prospective two-step FML packages, the autonomous baseline wins both comparisons; mean autonomous-minus-human delta is -0.185672 with SEM 0.164152. This result strengthens the negative average-performance evidence while leaving the high-tail taste/insight hypothesis open for future larger-budget studies.

We then extended the prospective package runner beyond the repeated Causality workspace to the other FML-bench workspace available on ubuntu-heshi, Fairness_fairlearn. This package is also negative for the co-pilot path, but for a different reason: both co-pilot branch candidates failed validation, so the logged frontier gate selected `abort_no_valid_branch` rather than forcing a continuation. The matched autonomous run completed with test primary metric 0.172152 (`abs_demographic_parity_diff_mean`, lower is better). The generated package audit now finds four passing prospective packages, all with complete attention-cost and taste/insight gate records: one controlled micro-task win, two negative Causality FML packages, and one Fairness FML invalid-continuation case. This broadens the benchmark shape but makes the average-performance story more conservative, not stronger.

4.4 Non-FML Benchmark and Program-Search Probe

Following the benchmark-selection principle above, we also used MLAGentBench as a non-FML evaluation source. We cloned MLAGentBench on the Ubuntu host and first ran its lightweight vectorization task with the built-in Agent baseline, which simply executes the starter `train.py` and submits the result. The official baseline completed with final score 3.172504 seconds and total benchmark time 3.365531 seconds, with no reported error flags.

We then wrapped the same task in a stricter local evaluator: before accepting a runtime, the evaluator compares the candidate `Conv2DLayer.forward` output against a nested-loop reference on a deterministic small input. Under this controlled evaluator, the starter program had median runtime 3.261186 seconds. A direct DeepSeek `deepseek-chat` rewrite produced plausible vectorized code but failed the correctness gate due to a numerical mismatch. In contrast, a three-iteration OpenEvolve-style run using the same DeepSeek model found a correct candidate at iteration 1 with median runtime 0.051882 seconds, a 62.86x speedup over the controlled starter program. This result supports the narrower claim that the program-search-escalation node can be valuable on machine-gradeable subproblems beyond FML-bench. It does not yet prove that the full co-pilot architecture improves whole-paper quality.

To test robustness, we repeated the same three-iteration OpenEvolve setup with additional random seeds. Across seeds 0, 1, 2, 3, 4, 7, 42, and 123, all eight runs retained a correct best program and improved over the controlled starter. The median best runtime was 0.024581 seconds, corresponding to a median speedup of about 132.67x over the starter runtime. Six of eight seeds found a sub-0.1-second program. The result is still not deterministic under tiny budgets: seed 7 only achieved a weak 1.09x speedup and seed 1 achieved a 15.57x speedup. This strengthens the evidence that the program-search module can find useful code transformations, while preserving the important caveat that search quality varies substantially by seed and budget.

To avoid making the non-FML evidence only about low-level runtime optimization, we added a second controlled probe using the built-in sklearn diabetes tabular regression dataset. This probe does not require Kaggle credentials and should not be reported as an official MLAGentBench score. It starts from a deliberately rudimentary mean predictor with mean RMSE 78.572189 across five deterministic splits. A direct DeepSeek rewrite recovered a standard Ridge-style baseline with mean RMSE 55.895460. Three 3-iteration OpenEvolve seeds also improved strongly over the mean predictor, with best RMSEs 55.895460, 55.946535, and 55.895460. The median OpenEvolve RMSE was 55.895460, matching the direct rewrite. This result broadens the benchmark coverage to a tabular modeling subproblem, but it also gives an important boundary condition: when the improvement is a standard small modeling change, direct editing can be as effective as program search. Thus the program-search gate should be selective, not automatic.

We also attempted three benchmark-expansion probes. First, we tried to add a second official MLAGentBench debug task, which maps to CIFAR10. The setup probe repaired a missing torchvision dependency by installing the matching CPU wheel for the local torch version, but the run stopped during dataset preparation because the 170 MB CIFAR10 archive was downloading too slowly for the interactive budget. Second, we probed ScienceAgentBench. The repository was present on the Ubuntu host, and its README points to the April 2026 verified split and benchmark_verified.zip, but the local benchmark directory did not contain the verified artifacts and the HuggingFace metadata request failed with [Errno 101] Network is unreachable. Third, we probed the official MLAGentBench imdb folder. After installing the missing datasets dependency required by its official eval.py, even a five-example dataset load failed with the same HuggingFace network error. We therefore report these as setup artifacts and not as benchmark scores. The next official non-FML score requires pre-cached data or a different network path, not a convenient retreat back to FML-bench alone.

4.5 Claim Audit

We ran a claim-evidence audit after the pilot experiments. A Monica-routed gpt-4o-mini review classified the core architecture contribution as reasonable but warned that the empirical evidence does not yet support broad claims that human gates improve paper quality or that the full co-pilot system outperforms autonomous AI Scientist-v2. We therefore treat those as hypotheses for the evaluation protocol rather than as conclusions. The audit supports only the narrower empirical claims reported above: OpenEvolve-style search can help on some machine-gradeable subproblems but is seed-sensitive under tiny budgets, direct editing can match OpenEvolve on a simple tabular modeling probe, branch-gate insertion is feasible in AI Scientist-v2-style logs, evaluator gates must reject non-executable and metric-gaming branches before continuation, and the first two matched Causality pairs give mixed evidence rather than a reliable human-gating advantage. A first attempt to extend online branch gating to Fairness_fairlearn produced two validation failures rather than a score, so we archive it as failure-mode evidence rather than matched-budget performance evidence. A follow-up evaluator-gate repair probe made the lesson sharper: an API-repaired fairness candidate was executable but worse on demographic parity difference (test 0.317603 versus the baseline 0.173030), while a degenerate all-negative predictor achieved a perfect target metric of 0.000000 but collapsed balanced accuracy to 0.500000. For fairness tasks, the gate therefore needs both an execution check and a utility floor before accepting a branch.

We also ran a refreshed paper-quality review through two Monica-routed reviewer models after adding the prospective matched-budget metric summary. gpt-4o-mini gave a weak-accept recommendation with scores of 4/5 for novelty and reproducibility but 3/5 for rigor and evidence. claude-3-7-sonnet-latest was stricter and gave a reject recommendation for a strong ML/NLP systems venue, mainly because the central human-gating and paper-quality claims remain unsupported, the strongest FML-bench prospective package is negative for co-pilot performance, and the system still lacks a paper-generating end-to-end comparison. Both reviewers converged on the same required next step: run autonomous AI Scientist-v2 and human-gated variants under equal budgets across more tasks and seeds, measure human attention cost, and separate completed system evidence from future plans more sharply.

5. Current Contributions and Unproven Claims

The current contributions are:

1. A modular architecture for collaborative automated research.
2. A formal schema for human intervention nodes in research agents.
3. A live Monica-routed hypothesis-frontier smoke that generates four new research-frontier candidates, critiques/ranks them, and selects frontier_004 for possible next-budget evaluation.
4. A same-model autonomous hypothesis-front-end baseline probe that compares the archived IGRE frontier portfolio with an autonomous AI Scientist-v2-style portfolio and scores IGRE 4 vs. autonomous 3 as front-end planning evidence only.
5. An offline expert-review taste-prior data probe over Hugging Face nhop/OpenReview, using streaming access to sample 160 rows from a 34,638-row expert-review corpus and verifying that it can support limited scientific taste-prior experiments.
6. A same-manuscript structured-feedback probe that operationalizes frontier_004 and compares informal feedback with IGRE-structured feedback through two revisions and a fixed-rubric model score.
7. A retrospective full-gate trajectory showing idea, evaluator, branch, program-search, and claim-audit gates serialized under the shared schema.
8. A rerunnable full-gate trace script that recomputes the gate chain from

archived experiment summaries while marking the output as artifact replay.

9. A first online full-gate smoke trajectory that exercises idea, evaluator, branch, program-search, and claim gates in one remote run, while producing a negative continuation outcome.

10. An evaluation protocol for measuring both mean benchmark performance and high-tail scientific upside under human intervention.

11. Initial remote OpenEvolve and FML-bench probes showing that the verifiable micro-evolution and frontier-steering operators can run on the Ubuntu host.

12. Two matched-budget FML-bench Causality comparisons between a human-gated branch continuation and a four-step autonomous AI Scientist-v2 baseline, with mixed outcomes.

13. A same-FML-step autonomous baseline for the first online full-gate smoke, showing a negative performance result for the human-gated continuation.

14. Non-FML program-search probes for runtime optimization and tabular regression, broadening benchmark coverage beyond FML-bench.

15. A reusable Codex skill for running the workflow.

16. Bilingual paper, usage artifacts, and claim-audit artifacts for reproducibility.

17. A Monica-routed paper-quality review artifact that records external model criticism before the next revision.

18. A human-gate attention-cost audit over 39 gates showing 1 complete operator-recorded attention-cost event and 38 incomplete records; future prospective experiment gates must record these fields before making attention-efficiency claims.

19. A taste/insight coverage audit showing 2 complete scientific-taste prior records and 37 gates that still lack complete taste/insight fields.

20. A prospective matched-budget package validator that defines the minimum non-synthetic evidence shape required before claiming paper-quality gains, human-attention efficiency, or superiority over autonomous AI Scientist-v2.

21. A controlled prospective Max-Cut micro-pilot package that passes this validator, with complete attention/taste logging, matched baseline metrics, claim audit, and same-run manuscript artifact.

22. Two prospective FML-bench Causality packages with complete attention/taste logging and matched autonomous baselines, both yielding negative co-pilot performance results in the small two-step setting.

23. A prospective package metric summary that separates passing audit packages by task, metric direction, co-pilot score, autonomous score, and claim implication; the current result is one positive controlled micro-task, two negative Causality FML-bench packages, and one Fairness_fairlearn invalid continuation package.

24. A matched mini-manuscript quality probe for the FML package, where Monica-routed gpt-4o-mini and claude-3-7-sonnet-latest both prefer the co-pilot package mini-manuscript over a generated autonomous mini-manuscript, with overall scores of 4 versus 3.

25. A matched full-manuscript generation probe that renders the same archived

FML evidence into two complete paper-shaped manuscripts and scores them with a deterministic internal rubric; the co-pilot manuscript scores 4.18 overall for structure, grounding, calibration, and method distinctness, while the autonomous manuscript scores 4.11 and is the only variant with a valid scalar FML test metric in the Fairness package.

26. A repeated same-continuous-trajectory paired online full-gate manuscript-production summary over three smoke runs, including one Fairness_fairlearn no-valid-branch failure trajectory; the valid Causality benchmark aggregate has 0 co-pilot wins, 1 autonomous win, and 1 tie, while Monica-routed model-review probes prefer the co-pilot manuscript in 6/6 reviewer calls.

27. A derived Human Co-Pilot Trace Dataset protocol that positions the author's real Codex sessions as a single-author longitudinal process corpus after privacy-preserving metadata extraction.

The current evidence does not yet prove that human gates improve paper quality or that the full co-pilot system outperforms autonomous AI Scientist-v2. The structured-feedback probe is a measurement artifact for one manuscript, not a replacement for independent expert review. Those stronger claims remain targets for the next benchmark stage.

6. Limitations

This proposal does not assume that human involvement always helps. Human gates can introduce bias, slow search, reduce exploration, and in short-budget benchmarks they may underperform a fully autonomous search policy. That is not a side note; it is part of the method's intended evaluation. IGRE treats human scientists as high-variance search operators whose value may appear in the upper tail rather than the mean: better problem taste, a more revealing evaluator, a sharper interpretation of failure, or a willingness to pursue a riskier but more original direction. Programmatic search can likewise optimize local metrics without improving scientific contribution, and under tiny budgets it may not outperform direct LLM editing. Expert paper ratings are costly and may vary across reviewers. The first version should make narrow claims and report negative results when gates fail to improve outcomes. The current selected-branch continuation evidence is mixed across two matched pairs and is not yet a statistically controlled benchmark. The retrospective full-gate trajectory and executable artifact replay show the schema, decision chain, and reproducible traversal logic. The online smoke trajectory does exercise all five gates in one remote run, but it uses a tiny budget, mixes an FML branch task with a knapsack program-search subproblem, and produced a worse continuation test score. A same-FML-step autonomous baseline also outperformed the human-gated continuation. Stronger claims require more tasks, more seeds, richer budget schedules, larger online trajectories, independent paper-quality review, and metrics that capture rare high-quality research outcomes rather than only average task score.

The current human-gate logs also lack measured attention cost. We can count decision artifacts, but cannot yet compute active review minutes or wall-clock latency. This prevents any claim that the proposed gates improve the ratio of research quality to human effort. The next matched-budget experiments must record attention cost prospectively.

The new prospective matched-budget package audit now passes on one controlled micro-pilot, two FML-bench Causality pilots, and one FML-bench Fairness pilot. This is progress in evidence shape, not in final empirical strength. The FML packages are negative or invalid-continuation cases for co-pilot average performance at this budget, and the packages do not provide independent expert evaluation of full paper quality. The paper therefore remains at pilot-system evidence until larger prospective experiments exist.

The derived Human Co-Pilot Trace Dataset improves ecological validity because it comes from the author's real Codex workflow, but it is still a single-author longitudinal process corpus. It can support process and case-study claims; it cannot by itself support population-level claims about scientists in general. The release audit reduces privacy and leakage risk by checking for secret-like strings and raw-log marker fields, but it is not a substitute for institutional human-subject review if the dataset is later expanded to multiple researchers. The OpenReview/Hugging Face expert-review probe is therefore complementary rather than substitutive: it offers a scalable offline proxy for scientific taste and reviewer judgment, but it does not contain real-time human decisions inside a co-pilot automated-research loop. A true online human-guided automated-science dataset would require a broadly deployed skill or research assistant used by many scientists, with consent, de-identification, artifact linking, and prospective logging of interventions, attention cost, and downstream outcomes. At present, this is more likely to be feasible for major agent or foundation-model platforms than for a single-author pilot project, so we treat it as a central future-work direction rather than as evidence already available in this paper.

The full-manuscript probe reduces one specific gap but does not close the top-conference evidence gap. It shows that archived FML packages contain enough structured evidence to generate two complete, claim-calibrated manuscripts under a matched prompt-free template. The latest Fairness probe scores the co-pilot manuscript 4.18 and the autonomous manuscript 4.11 on the internal rubric, but the autonomous manuscript is the only variant with a valid scalar FML test result. The repeated same-continuous online smoke summary now shows that fresh online co-pilot trajectories can produce complete manuscript-shaped artifacts and same-run autonomous manuscript comparators. Across the two valid Causality paired smokes, benchmark outcomes are 0 co-pilot wins, 1 autonomous win, and 1 tie, with mean test MAE 0.754120 for co-pilot and 0.643337 for autonomous. The third Fairness_fairlearn paired smoke is a no-valid-branch failure trajectory rather than a scored performance comparison. Monica-routed model reviewers prefer the co-pilot manuscripts in 6/6 calls, but this is a measurement-readiness signal rather than expert review. This is exactly why the co-pilot claim must remain about method distinctness, scientific taste logging, and high-tail research search rather than average short-budget benchmark superiority.

The taste/insight evidence is also only at the logging-readiness stage. The archive now contains two complete scientific-taste prior records: one grounded in the author's instruction to broaden benchmarks and foreground high-tail research taste, and one operator-recorded decision to prioritize attention/taste measurement over more weak benchmark runs. Neither record shows that the decision improved downstream research outcomes. Future runs must log taste rationales prospectively rather than reconstructing them from successful outcomes.

The current implementation now has smoke-level repeated same-continuous- trajectory paper-generating comparisons, but it is still too small for a systems-paper claim. The next systems-paper draft must report larger matched trajectory pairs across tasks and seeds, from hypothesis generation to final claim-audited manuscripts.

7. Conclusion

Co-Pilot AI Scientist v3 reframes automated scientific discovery as insight-gated research evolution. The system preserves the strengths of autonomous agents while giving human scientists explicit, logged, and experimentally testable points of influence. Its strongest claim is not that humans always improve average benchmark performance. The deeper claim is that human scientific taste and insight can reshape the search distribution toward rarer, more original, and potentially field-opening outcomes. If future matched benchmarks support this high-tail hypothesis, co-pilot research systems may produce better papers not by removing humans from science, but by using human attention where it can change what kind of science is attempted. The present paper should be read as a reproducible system proposal with pilot evidence, not as a final proof of superiority.